

Products and Inventory Management

Single source of truth for product data that AI, website, ads, orders, and courier use.

Why this exists

Bad product data makes AI lie and orders fail. Inventory must be simple but strict.

Primary user	Owner, product manager, staff.
Business value	Keeps AI useful, reduces overselling, enables ads/catalog and repeat orders.
Build priority	P0 - AI depends on it.

1. Page Goal

- Add and manage products, variants, stock, images, pricing, and delivery details.
- Support bulk import/export for sellers moving from spreadsheets.
- Track low stock and reserved stock from pending orders.
- Give AI approved product facts only.

2. How It Should Look

Products table: image, name, price, stock, status, category, actions.
Product editor: left fields, right live product card preview.
Inventory view: stock in hand, reserved, sold, returned, low stock.
Bulk import modal: upload, map columns, validate, confirm.

3. Main Sections

Product list	Search, filter, status, bulk update.
Product editor	Core info, variants, images, stock, SEO, AI notes.
Inventory ledger	Stock change history with reason and staff name.
Categories	Website navigation and AI recommendation grouping.
Import/export	Excel/CSV in and out.

4. User Actions

Add product	Creates product draft.	Cannot publish without name and price.
Update stock	Changes stock and writes ledger entry.	Require reason for negative adjustment.
Bulk edit	Updates selected products.	Preview changes before apply.

Mark inactive	Hides product from store and AI.	Existing orders still keep product snapshot.
Import file	Validates rows and creates/updates items.	Duplicate SKU asks merge or create new.

5. Data Fields

- Product: name, SKU, category, description, price, compare price, cost optional.
- Inventory: stock, reserved, low stock threshold, warehouse optional V2.
- Variants: size, color, material, image, price override, stock.
- AI facts: selling points, who it is for, care instructions, policy exceptions.

6. AI / Automation Behavior

- AI answers only from active products and stock.
- AI recommends alternatives when requested item is out of stock.
- AI flags missing product fields that weaken replies.
- AI can create draft descriptions from image/name, owner approves.

7. Empty, Error, and Safety States

- Empty state must show one clear next button, not a blank table.
- Every destructive action needs confirm: delete, cancel order, disconnect account, refund mark.
- If third-party API fails, keep local draft and show retry.
- AI must never invent price, stock, delivery charge, delivery date, or policy.

8. Acceptance Checklist

- Stock never goes below zero unless seller enables backorder.
- Bulk import gives clear row-level errors.
- Order creation reserves stock; cancellation releases stock.
- Inventory ledger shows who changed what and when.

9. Do Not Build Yet

- No complex manufacturing/BOM inventory.
- No multi-warehouse in V1 unless one seller truly needs it.
- No barcode system until offline warehouse flow exists.